SWYNEX AI Problem Design

Project Title

Student Feedback Sentiment Classification Using AI

1. Problem Statement

Students provide feedback about their classes and teachers. Reading all feedback manually can take time. This project proposes an AI-based solution to classify student feedback into three categories: Positive, Negative, and Neutral.

2. Target Users

The target users are teachers, schools, and educational institutions that want to understand student feedback more efficiently.

3. Data Source

A small sample dataset will be created manually using example student feedback sentences. Each sentence will have a sentiment label. No personal student information will be collected.

4. Proposed AI Approach

The proposed system will use Natural Language Processing (NLP) and text classification to categorize feedback as Positive, Negative, or Neutral.

5. *Constraints*

- The dataset may be small.
- Spelling mistakes may affect classification.
- Sarcasm and complex sentences may be difficult to understand.
- The model may make incorrect predictions.
- Human review will remain important.

6. Success Criteria

- Classify feedback into three sentiment categories.
- Aim for at least 80% accuracy on a separate test dataset.
- Evaluate precision, recall, and F1-score.
- Record the actual results after testing the model.

The 80% accuracy is a target, not a verified result.

7. Evaluation Approach

The dataset will be divided into training and testing sets. The system will be tested on unseen feedback. Accuracy, precision, recall, and F1-score will be used to evaluate performance.

8. Expected Outcome

The proposed system will help organize student feedback into meaningful categories and demonstrate a practical application of AI in education.

9. Ethical Considerations

The project will use sample feedback without personal information. AI predictions will be treated as estimates rather than absolute judgments.

10. Conclusion

This project defines a practical AI problem in education and explains its users, data source, proposed approach, constraints, and success criteria.